

A138 – Anomalous Magnetic Moments: Ratios Exact, Absolute Values with Fractal Correction

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.1 Purpose

[STIPULATION] This document treats the anomalous magnetic moments of the leptons ($g-2$) in the FFGFT framework. The central distinction: *ratios* of $g-2$ differences are exactly derivable from the fractal structure, because all conversion factors cancel. *Absolute values* in SI units depend on the projection factor k_{geom} and on SI bridge constants – they have, after correction by $K_{\text{frak}}^{3/2}$, an accuracy of 0.01–0.15 %, but the absolute value itself is not a parameter-free geometric prediction. That is the structurally honest status.

.2 Physical Foundations

[B] The anomalous magnetic moment is the deviation from the Dirac value:

$$a_\ell = \frac{g - 2}{2}.$$

For a point particle Dirac predicts $g = 2$, hence $a = 0$. Quantum effects (vacuum polarisation, vertex corrections) generate $a \neq 0$.

[B] In FFGFT leptons are winding structures in the 4D torsion lattice; the three generations carry different fractal Hausdorff exponents: electron (simple winding, 1st generation), muon (fractally branched with exponent $p_\mu = 5/3$), tau (more branched, $p_\tau = 4/3$). The anomalous moment follows from rotation of the winding, charge distribution on it, and projection $4D \rightarrow 3D$.

.3 The Geometric Formulas for Absolute Values

[K] The fundamental parameters of the FFGFT g -2 calculation:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618 \dots \quad (\text{golden ratio}), \quad f = 7500 \quad (\text{sub-Planck factor}), \quad \xi = \frac{4}{3} \times 10^{-4}.$$

[K] Electron. From the 3D surface of the 4D winding $S_3 = 2\pi^2 = 19.739$ and the geometric projection factor

$$k_{\text{geom}} = \frac{2}{\sqrt{\varphi}} \times \sqrt{2} = 2.224$$

follows the ideal formula:

$$a_e^{(\text{ideal})} = \frac{S_3/f}{k_{\text{geom}}} = \frac{19.739/7500}{2.224} = 1.184 \times 10^{-3}.$$

Experiment: $a_e = 1.1597 \times 10^{-3}$. Deviation ideal: 2.03 %.

[K] Muon. The fractal additional winding:

$$\Delta a_{\text{fractal}} = \frac{4\pi}{f^{p_\mu}} = \frac{4\pi}{7500^{5/3}} = 4.373 \times 10^{-6}, \quad a_\mu^{(\text{ideal})} = a_e + \Delta a_{\text{fractal}} = 1.188 \times 10^{-3}.$$

Experiment: $a_\mu = 1.1659 \times 10^{-3}$. Deviation ideal: 1.89 %.

.4 Resolution of the Systematic Deviation: $K_{\text{frak}}^{3/2}$

[K] The $\sim 2\%$ deviation of the ideal formulas is not an open question but is explained by the fractal correction. The effective projection factor:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.224}{(1 - 100\xi)^{3/2}} = \frac{2.224}{0.98007} = 2.2692.$$

The exponent $3/2 = D/2$ describes how the fractal structure acts on the projection across three spatial dimensions. $K_{\text{frak}} = 1 - 100\xi$ is the same correction as in A040 – no new parameter.

[K] With the corrected factor:

Lepton	T0 ideal	T0 corrected	Experiment	Dev. ideal	Dev. corr.
Electron	1.184×10^{-3}	1.1598×10^{-3}	1.1597×10^{-3}	2.03 %	0.01 %
Muon	1.188×10^{-3}	1.1642×10^{-3}	1.1659×10^{-3}	1.89 %	0.15 %
Tau	1.269×10^{-3}	1.2454×10^{-3}	(not measured)	–	Prediction

[B] Methodological clarification. The corrected absolute value is a consequence of the FFGFT-internally consistent K_{frak} – not a fit. The correction agrees with the experimentally reconstructed value $k_{\text{rec}} = S_3/(f \cdot a_e^{\text{exp}}) = 2.2696$ to 0.014 %. But: k_{geom} and f depend on SI bridge constants and measured quantities. An absolute SI value without any measured anchor – as the SM can achieve with α – is not available at the same level in FFGFT (the same holds for masses, A130, A080).

.5 Ratios: Parameter-Free and Exact

[B] In contrast to absolute values, in ratios of g -2 differences all conversion factors cancel:

$$\frac{\Delta a(\tau - \mu)}{\Delta a(\mu - e)} = \frac{4\pi/f^{4/3} - 4\pi/f^{5/3}}{4\pi/f^{5/3}} = \frac{f^{5/3}}{f^{4/3}} - 1 = f^{1/3} - 1.$$

Numerically: $f^{1/3} = 7500^{1/3} = 19.57$, hence

$$\frac{\Delta a(\tau - \mu)}{\Delta a(\mu - e)} = f^{1/3} - 1 = 18.57$$

No k_{geom} , no K_{frak} , no SI factor, no α . The ratio depends exclusively on $f = 7500$ – and f is the sub-Planck factor $30000/4$, following directly from $\xi = 4/30000$.

[S] Structural parallel to the Koide formula. The Koide formula $(m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 \approx 2/3$ is confirmed to 0.001 % because there ratios rather than absolute values are compared. The same structure operates here: the exponent step $\Delta p = 1/3$ of the masses (A100) appears directly as $q_\mu = 5/3$, $q_\tau = 4/3$, $\Delta q = 1/3$ in the g -2 exponents. No coincidence – the same geometric origin.

.6 The Bridge Formula: Masses and g -2 Connected

[B] From the mass formulas: $m_\tau/m_\mu = (125/144) \cdot f^{1/3}$. From the g -2 formulas: $\Delta a(\tau - e)/\Delta a(\mu - e) = f^{1/3}$. Combining gives the bridge formula:

$$\frac{\Delta a(\tau - e)}{\Delta a(\mu - e)} = \frac{144}{125} \cdot \frac{m_\tau}{m_\mu}$$

The parameter f cancels. No α , no k_{eff} , no K_{frak} . The prediction follows solely from the shared $1/3$ step size of masses and g -2.

[K] Tau prediction from the bridge formula. With measured values $a_e^{\text{exp}} = 1.160 \times 10^{-3}$, $\Delta a(\mu - e)^{\text{exp}} = 6.269 \times 10^{-6}$, $m_\tau/m_\mu = 16.817$:

$$\begin{aligned} \Delta a(\tau - e) &= \frac{144}{125} \cdot 16.817 \cdot 6.269 \times 10^{-6} = 1.214 \times 10^{-4}, \\ a_\tau^{(\text{bridge})} &= a_e + \Delta a(\tau - e) = \boxed{1.282 \times 10^{-3}}. \end{aligned}$$

The SM calculates $a_\tau^{\text{SM}} = 1.177 \times 10^{-3}$. Difference: 9 %. Testable at Belle II (~ 2027).

.7 Fair Comparison: SM and FFGFT

[S] The SM circularity. The SM extracts α from the a_e measurement and then computes a_e back. For electron and muon this is a circular fit – a precision calculation, not an independent prediction. FFGFT derives α from ξ alone: $\alpha = \xi \cdot E_0^2$ with $E_0 = 7.398 \text{ MeV}$ gives $1/\alpha_{\text{T0}} = 137.035 - 0.0005 \%$ from experiment (A130). That is a genuine, independent prediction, though at a different precision level than the SM perturbation calculation.

[S] Where the theories diverge: Not at e or μ (both theories lie close to experiment there), but at the τ . T0: $a_\tau \approx 1.282 \times 10^{-3}$, SM: 1.177×10^{-3} – factor 7 larger deviation than the muon discrepancy. That is the decisive test.

.8 Verification Script

- Verification of all formulas: ideal and corrected absolute values, ratio $f^{1/3} - 1$, bridge formula with experimental values, tau prediction: `python/A_Serie_Skripte/a138_g2.py`

.9 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$a_\ell = (g - 2)/2$	Anomalous magnetic moment of lepton ℓ
$f = 30000/4 = 7500$	Sub-Planck factor; follows directly from $\xi = 4/30000$
$k_{\text{geom}} = (2/\sqrt{\varphi})\sqrt{2}$	Geometric projection factor
$k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$	Effective projection factor after fractal correction
$S_3 = 2\pi^2$	Surface of the 3-sphere (volume of the 4D unit ball)
$p_\mu = 5/3, p_\tau = 4/3$	Fractal Hausdorff exponents of the generation structure

.10 Open Questions

The absolute SI values depend on SI bridge constants and measured quantities. The correction $K_{\text{frak}}^{3/2}$ explains the systematic deviation; but the geometric projection factor k_{geom} is not derived at the same level as the pure ratios.

The tau prediction ($a_\tau \approx 1.282 \times 10^{-3}$) has not yet been measured. Belle II will test it. Until then it is a prognosis – precisely formulated, but not a result.

The exponent $3/2$ for the fractal correction is geometrically motivated ($D/2 = 3/2$) but not uniquely proved. That exactly $D/2$ and not some other exponent appears is a consistency observation, not a derivation.

.11 Supersedes

This document subsumes: Dok. 018 (all three versions 018-10/11/12, anomalous magnetic moments of the leptons: geometric formulas, $K_{\text{frak}}^{3/2}$ correction, ratio predictions, bridge formula masses $\leftrightarrow g-2$, tau prediction, fair SM comparison). Not taken over: narrative passages and cosmological generalisations from Dok. 018.

.12 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.